

Tracing Functional Correctness in DLMs with Sparse Autoencoders

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Abstract

Prior probing of diffusion language models (DLMs) suggests that correctness signal grows monotonically through denoising. Using DLM-Scope sparse autoencoders, we instead trace the signal feature by feature across two DLMs and four tasks. Sparse fail-vs-pass concentration appears in task-dependent phases: code and structural tasks form a broad mid-denoising plateau, while reasoning tasks emerge only at the final state. On LLaDA-8B / MBPP, a dense step sweep across seeds confirms the plateau spans steps 48–116 with multiple sub-steps reaching $p < 0.05$ and a within-plateau hand-off from one persistent feature to another at the late phase. The peak phase agrees across LLaDA-8B and Dream-7B on 3/4 tasks, although most individual cross-grid cells do not reach per-step significance. However, single- and multi-feature steering fails to flip correctness in a small pilot intervention set, suggesting these sparse features are diagnostic under our current protocol rather than directly controllable knobs.

1 Introduction

Diffusion language models (DLMs) generate text by iteratively denoising a fully masked sequence over ~ 100 steps (Austin et al., 2021a; Sahoo et al., 2024; Lou et al., 2024; Nie et al., 2025; Ye et al., 2025). Chiu (2026) shows that simple classifiers trained on hidden states gain 0.08–0.11 AUC predicting functional correctness as denoising proceeds, framing the process as a monotonic sharpening of an aggregate scalar. This view leaves a basic question unanswered: *which* representational components carry the failure signal, and *when* during denoising do they form?

Contributions.

- **Mid-denoising plateau.** LLaDA-8B / MBPP has a broad plateau over steps 48–116 (3-seed dense sweep), with f15601 top-ranked at the early plateau and f3892 at the late plateau.

- **Phase agreement.** Peak phase agrees across LLaDA-8B and Dream-7B on 3/4 tasks, while per-cell significance is concentrated in the strongest LLaDA-MBPP condition.
- **Steering null.** Single-, top-5, and reverse feature interventions fail to flip correctness in a pilot intervention sweep.

Code and figures will be released on acceptance.

2 Related Work

Discrete diffusion and masked DLMs generate by iteratively denoising masked tokens rather than left-to-right continuation (Austin et al., 2021a; Sahoo et al., 2024; Lou et al., 2024; Nie et al., 2025; Ye et al., 2025), and prior probing work on the same models and tasks studies aggregate hidden-state predictability and reports monotonic gains in correctness AUC through denoising. SAEs decompose residual streams into sparse named features and have been used primarily to interpret autoregressive LMs (Bricken et al., 2023; Cunningham et al., 2024; Gao et al., 2024); DLM-Scope extends this toolkit to DLMs with Top- K SAEs (Makhzani and Frey, 2014; Gao et al., 2024) trained on masked-position residuals across mask rates $t \sim U(0, 1)$ (*Mask-SAE*), so the full denoising trajectory is in distribution (Wang et al., 2026). Recent AR SAE studies report both successful steering for refusal and instruction following (O’Brien et al., 2024; He et al., 2025) and cautions that contrastive features may track lexical correlates or that steering degrades unrelated capabilities (Galichin et al., 2025; Korznikov et al., 2025). We reuse the DLM-Scope SAEs to ask whether the scalar correctness trend on DLMs decomposes into temporally localised sparse features, treating the steering question as open rather than presumed causal.

3 Method

Models and SAEs. LLaDA-8B (Nie et al., 2025) (block-wise unmasking, block length 32) and Dream-7B (Ye et al., 2025) (global linear schedule) both denoise a ≤ 256 -token generation region over $T=128$ steps conditioned on a fully visible prompt. We use the DLM-Scope L26 LLaDA-mask SAE and L23 Dream-mask SAE (trainer index 2, $d_{\text{sae}}=16,384$, $K=160$). The AR comparison (DLM-Scope L23 SAE on Qwen-2.5-7B (Team, 2024)) is a single-token reference point, not a trajectory: we encode the last-prompt-token residual once per sample as a single-step analog of the DLM’s pre-generation state.

Data. We reuse cached hidden states and functional-correctness labels from Chiu (2026): MBPP-sanitized (257) (Austin et al., 2021b), JSON schema (272), GSM8K (1,319) (Cobbe et al., 2021), and ARC-Challenge (1,172) (Clark et al., 2018), generated with seed 0 and $T=128$ steps. Hidden states are mean-pooled over four equal-length generation regions at seven checkpoint steps $\{0, 1, 4, 16, 32, 64, 127\}$.

Per-step SAE encoding. At each (model, step, layer) cell we take the region-mean hidden state $h \in \mathbb{R}^d$ per sample and encode it: $z = \text{TopK}(\text{ReLU}(W_{\text{enc}}(h - b_{\text{dec}}) + b_{\text{enc}})) \in \mathbb{R}^{d_{\text{sae}}}$, with $W_{\text{enc}} \in \mathbb{R}^{d_{\text{sae}} \times d}$, $W_{\text{dec}} \in \mathbb{R}^{d \times d_{\text{sae}}}$, $b_{\text{enc}} \in \mathbb{R}^{d_{\text{sae}}}$, $b_{\text{dec}} \in \mathbb{R}^d$, and TopK keeps the $K=160$ largest entries. Feature i is *active* on a sample iff $z_i > 0$.

Fail-vs-pass enrichment. Let $\mathcal{F}, \mathcal{P}$ be the fail and pass sets at a cell. Feature i ’s enrichment is the difference of active rates, $\text{enr}(i) = \frac{1}{|\mathcal{F}|} \sum_{s \in \mathcal{F}} \mathbb{K}[z_i^{(s)} > 0] - \frac{1}{|\mathcal{P}|} \sum_{s \in \mathcal{P}} \mathbb{K}[z_i^{(s)} > 0]$. We use the top-20 most fail-enriched features as the analysis subset per cell.

Cluster discriminability with permutation null. We run KMeans on the fail samples for $K \in \{2, 3, 4, 5\}$ using the top-20 features (continuous activations) and report the silhouette of the best K . The permutation null shuffles labels $1000\times$, re-selects the top-20 features on each shuffled label set, then recomputes silhouette, controlling for the post-hoc feature selection. We report $p = \Pr[\text{silhouette}_{\text{perm}} \geq \text{silhouette}_{\text{obs}}]$. The silhouette is computed on fails only and measures substructure within failures, not fail-vs-pass separability directly. We therefore interpret it as *concentration of failure modes* in a selected sparse subspace,

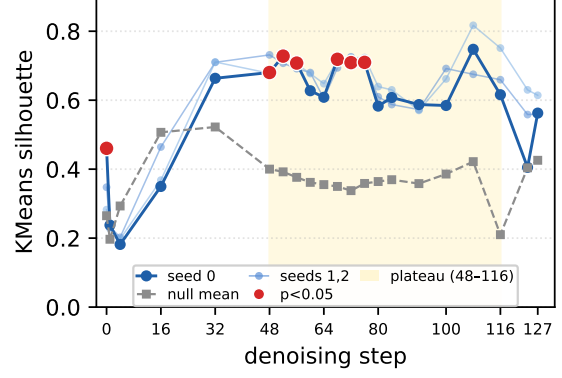


Figure 1: LLaDA-8B / MBPP: KMeans silhouette on the top-20 fail-enriched features vs. permutation null mean across denoising steps. All 21 checkpoints (anchors $\{0, 1, 4, 16, 32\}$, plateau $\{48-80\}$, post-plateau $\{84-124\}$, final 127) are from a single new-prompt generation run; faint lines overlay two additional generation seeds (Appendix I). Shaded region marks the mid-denoising plateau $[48, 116]$ where most steps reach $p < 0.05$; step 64 sits inside the plateau as a local dip, not a sharp peak.

and cross-check fail/pass separability with supervised AUC in the same subspace (Appendix G, per-fold selection).

Steering. We write *fail*→*pass* for a sample whose label flips from fail to pass after intervention, and *pass*→*fail* for the reverse pass-side regression. Given a target feature i^* and strength α , a forward hook on the SAE layer’s transformer block re-encodes the residual at every step $t \geq t_0$, reads z_{i^*} , and in-place subtracts $\alpha \cdot z_{i^*} \cdot W_{\text{dec}}[:, i^*]$. $\alpha < 0$ amplifies the feature (reverse intervention); for multi-feature steering we sum contributions over a set of indices. A zero-out smoke test (force $h \leftarrow 0$) confirms the hook propagates: it garbles trailing tokens but leaves the early-committed function body intact, consistent with LLaDA committing blocks 0–3 by step ~ 48 .

4 Results

Mid-denoising plateau. Figure 1 reports a 21-step trajectory for LLaDA-MBPP from a single generation run, with three seeds (Appendix I): anchors $\{0, 1, 4, 16, 32\}$, plateau dense $\{48, 52, \dots, 80\}$, post-plateau dense $\{84, 92, \dots, 124\}$, and final state 127. Rather than a sharp peak at step 64, the trajectory shows a broad plateau spanning steps 48–116: the signal-to-null gap stays in $[+0.19, +0.41]$ throughout this range and only collapses at

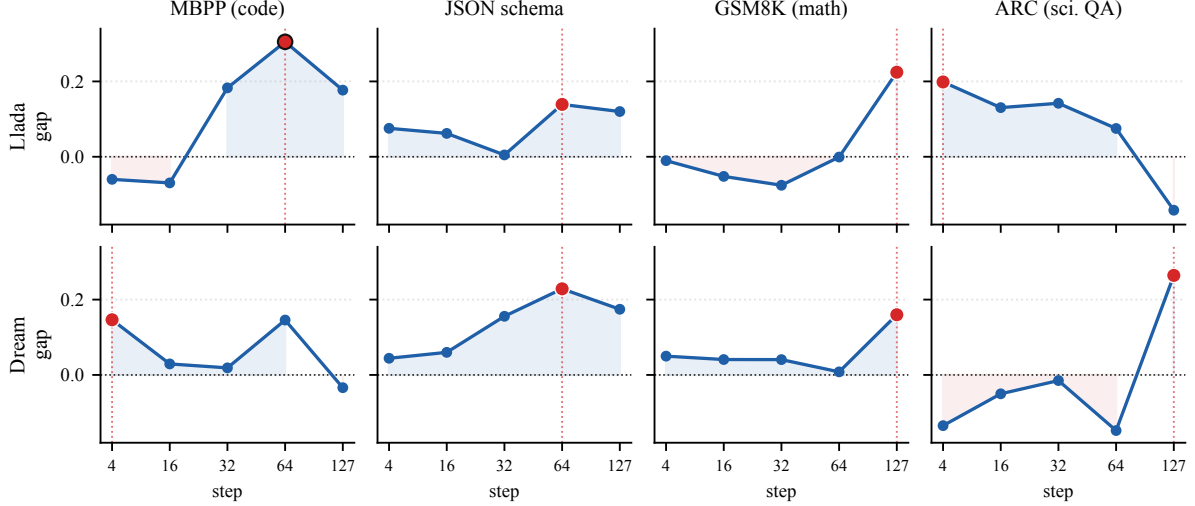


Figure 2: Cross-model $\times$ cross-task trajectory grid. Each panel plots signal-to-null gap (observed KMeans silhouette minus permutation null mean) across denoising steps for one (model, task) cell; red marker = peak gap step; black outline = $p < 0.05$ permutation significance. Code (MBPP) and structural (JSON) tasks peak at mid-denoising (step 64) on both DLMs. Math (GSM8K) and—on Dream—science QA (ARC) peak at late denoising (step 127). The peak step shows task-level agreement across LLaDA-8B and Dream-7B on three of four tasks; only ARC disagrees (LLaDA peak step 4, Dream step 127).

step 124 as the model approaches its final state. f15601 is the top-1 fail-enriched feature at 7/9 of the early-plateau dense steps and remains dominant through step 100; in the late-plateau steps 108–116 it is replaced by f3892 as the top marker, indicating a within-plateau hand-off rather than feature swapping outside the plateau. At steps 0–16 the top fail feature changes every step (Appendix A), consistent with the residual stream not yet encoding committed-token decisions when most positions are still masked. Chiu (2026) reports that aggregate raw-probe AUC *monotonically* increases over these same steps (for LLaDA / MBPP, from ~ 0.75 at step 0 to ~ 0.84 at step 64); the feature-level plateau is thus invisible to scalar probing.

Cross-model $\times$ cross-task phase agreement. Figure 2 reports the full 2×4 trajectory grid (full numeric table in Appendix E). Three archetypes emerge across cells. **Mid-denoising peak** (signal peaks at step 64): LLaDA-MBPP (+0.31), LLaDA-JSON (+0.14), Dream-JSON (+0.23), Dream-MBPP (+0.15). This is the shape of the persistent-feature trajectory described in §4: a fail-discriminative cluster crystallises during the middle of denoising and partially dissipates by step 127. **Late-denoising peak** (signal peaks at step 127): LLaDA-GSM8K (+0.22), Dream-GSM8K (+0.16), Dream-ARC (+0.26). Here the cluster

signal is weak or below null at mid-denoising and only emerges near the final state, consistent with reasoning tasks requiring more denoising before the failure signature crystallises. **Early-decline** (LLaDA-ARC peaks at step 4, +0.20): an outlier where the cluster appears only at the earliest steps and decays thereafter. The peak position shows *task-level agreement across DLMs*: three of four tasks (MBPP, JSON, GSM8K) have matching peak step across LLaDA and Dream; only ARC differs across models. At the feature level this mirrors the structural-vs-reasoning emergence split of Chiu (2026)—structural and code tasks form features mid-denoising; reasoning tasks form them late—now visible as a peak position rather than a continuous AUC curve. Appendix C summarises the 12 (DLM and AR) cells at step 64; across the full grid only two cells reach $p < 0.05$ at any single step (LLaDA-MBPP step 64, Qwen-JSON step 64). Thus the cross-DLM result should be read as a descriptive phase-alignment claim, with statistical support strongest for the LLaDA-MBPP plateau.

Top-feature drift and persistence. On LLaDA-8B / MBPP, pairwise Jaccard of the top-20 sets is high among steps 0–4 (≥ 0.48 , before committed-token information is available), moderate between 32 and 64 (0.29, the plateau regime), and falls to 0.14 between 64 and 127. On the sparse 7-step grid, only f15601 appears at all three peak-or-later

condition	from	fail→pass	pass→fail
baseline	none	0/8	0/3
suppress f15601	s64	0/8	0/3
suppress f15601	s16	0/8	0/3
suppress top-5	s64	0/8	0/3
reverse f15601	s64	0/8	0/3

Table 1: Pilot steering intervention summary on LLaDA-8B / MBPP; entries show flips / evaluated cases. Conditions: baseline (no intervention); suppress f15601 at step 64 (peak) or step 16 (pre-peak); suppress the top-5 fail features at step 64; reverse intervention ($\alpha = -5$) amplifying f15601 at step 64. No condition converts any fail→pass or pass→fail.

steps; the dense sweep (Appendix H) sharpens this by showing f15601 is the top-1 fail-enriched feature at 7 of 9 dense steps in [48, 80], supporting a persistent statistical marker rather than feature swapping inside the plateau. Outside the plateau, the sparse support reorganises as denoising commits different token blocks: early features are unstable, the compact failure marker f15601 crystallises during 48–76, and the final state at step 127 spreads signal over a different set of atoms. The temporal peak is therefore a concentration effect in sparse feature space, not merely a monotonic increase in correctness information. Full Jaccard matrix and per-feature enrichment trajectories are in Figure 3 (Appendix A).

Pilot steering null result. We test whether the cluster-level feature signature can be directly controlled, restricting to LLaDA-8B / MBPP where the cluster reaches significance. We run five intervention conditions on cluster-1 fail cases and pass-side regression controls (Table 1; entries report flips / evaluated cases). Suppressing the persistent peak feature f15601 ($\alpha=5$) at the peak window (step 64) or before formation onset (step 16) yields *zero* fail→pass conversions and *zero* pass→fail regressions. Multi-feature suppression of the top-5 fail features at step 64 is also null. Reverse intervention ($\alpha = -5$, amplifying f15601) is null on pass-side cases as well. The hook does propagate—a separate zero-out test ($h \leftarrow 0$ at $t \geq 0$) destroys trailing generation while leaving the function body intact, consistent with LLaDA committing blocks 0–3 by step ~ 48 . This pilot result does not rule out causal roles for these features, but it does show that correctness is not flipped by ablating one direction or a small set of directions under this intervention protocol.

5 Discussion

Statistical vs. causal features. f15601 fires on 65% of LLaDA-8B / MBPP fails vs. 27% of passes at step 64—a $\sim 6.6\sigma$ per-feature effect—yet ablating it (or the top-5 fail features) in any of four intervention windows flips no label. The cluster signature is useful as a diagnostic, but our intervention evidence is insufficient to treat this sparse direction as a knob that controls functional correctness. This contrasts with positive AR steering results on refusal and instruction following (O’Brien et al., 2024; He et al., 2025) but is consistent with reports that activation steering may degrade unrelated capabilities (Korznikov et al., 2025). More decisive causal tests would need paired counterfactual replacements, multi-layer interventions, or attribution from output-level correctness back to the SAE layer.

Why do peak times differ by task? By step 64, LLaDA has unmasked blocks 0–3 (the function signature and early body) but not blocks 4–7. For code and structural tasks, this mid-state appears to be where fail-vs-pass information is maximally factored into a small set of sparse features: enough tokens are settled to expose the failure mode, while later tokens still carry uncertainty. By step 127 the signal still exists but is less concentrated in the top-20 sparse features. For GSM8K and Dream-ARC, by contrast, the sparse cluster peaks only at the final checkpoint, suggesting that reasoning failures become feature-separable later than format or code failures. This task-dependent timing also explains why a single intervention window is unlikely to control correctness across tasks.

6 Conclusion

Tracing functional-correctness signal feature by feature reveals temporal structure that scalar probes miss: sparse failure markers form task-dependent phases. Code and structural tasks form a mid-denoising plateau, while reasoning tasks peak at the final state; this peak-phase pattern agrees across LLaDA-8B and Dream-7B on three of four tasks, with strongest support on LLaDA-MBPP. In that cell, a 3-seed dense sweep confirms a steps 48–116 plateau and a late hand-off from f15601 to f3892. Pilot linear-feature interventions do not flip correctness, so these sparse features are currently diagnostic markers; causal control likely requires stronger interventions.

Limitations

SAE coverage. All findings depend on a single DLM-Scope Mask-SAE per model (L26 for LLaDA, L23 for Dream), trained with mask rate $t \sim U(0, 1)$ on masked-position residuals. The full denoising trajectory is therefore in distribution, but features extracted at other layers, the Unmask-SAE trained on already-revealed positions, or higher-capacity dictionaries ($K > 160$) may surface different failure correlates. Early-step instability (steps 0–16) reflects the residual stream not yet encoding committed-token decisions rather than SAE distribution shift.

Feature semantics. We identify persistent dictionary atoms such as f15601 and f3892 by statistical enrichment and temporal persistence, not by human-validated semantic labels. The present experiments therefore support a claim about when sparse failure markers concentrate, but they do not name the underlying error modes captured by each feature. Manual cluster inspection or activation-maximising examples would be needed to turn these feature IDs into interpretable failure categories.

Permutation test interpretation. Our permutation null re-selects the top-20 fail features on each shuffled label set before recomputing silhouette. This is conservative (any post-hoc selection procedure that could find clusters on random labels is included in the null), and it inflates the null mean noticeably (e.g. 0.36 for LLaDA-8B / MBPP at step 64). A consequence is that only two of twelve cells reach $p < 0.05$ at an individual step on the sparse 7-grid even though the DLM grid shows systematic plateau structure. The dense sweep in Appendix H shows 7 of 9 sub-steps in [48, 76] reach $p < 0.05$ on LLaDA-MBPP, so sparse-grid per-step significance is an underestimate; we present our cross-model claims as plateau-shape claims rather than per-cell significance.

Steering coverage. Our intervention sweep tests four denoising windows, two feature-set sizes (single and top-5), and one magnitude pair ($\alpha \in \{+5, -5\}$) at a single SAE layer per DLM (L26 for LLaDA, L23 for Dream). We do not test multi-layer joint interventions, non-linear projections, or feature-selection strategies that use the per-step top-1 (where the top fail feature differs from f15601). Effect-size diagnostics (Appendix J) confirm the intervention reaches $\|\Delta h\|/\|h\| \in [4.4\%, 14.8\%]$ at

the SAE-layer residual, so the null result is not because the perturbation was silent; rather, a negative result on this protocol is narrower than a negative result on all possible linear-feature interventions.

Cluster K and choice of metric. We use silhouette on KMeans with $K \in \{2, 3, 4, 5\}$ and report the best K per cell. On LLaDA-8B / MBPP the best K is 2 at every step; on Dream / MBPP step 64 the best K is 3 with one small (size-26) sub-cluster. Appendix K reports top- N and clustering-method sensitivity: silhouette and significance are robust across $N \in \{10, 20, 30, 50\}$ at step 68 (the plateau peak) and across KMeans/Agglomerative/HDBSCAN. A labels-hidden $K=2$ test on the full fail+pass set yields $\text{ARI} \approx 0$, confirming the silhouette measures within-fail sub-structure rather than fail/pass separability; supervised AUC in Appendix G reports the latter.

Cross-model comparison. LLaDA uses block-wise unmasking with block length 32, whereas Dream uses a global linear schedule. At a matched checkpoint step, the spatial distribution of unmasked tokens differs between the models, so step-axis comparisons in Figure 2 reflect approximate denoising-progress alignments rather than identical model states. We do not attempt to map between block-wise commit fractions and global mask fractions on a per-sample basis. Our AR baseline (Qwen-2.5-7B) uses last-prompt-token hidden states; this is the natural single-token analog of the DLM’s pre-generation state but is not a temporal trajectory.

Task coverage. Functional correctness is verified across four tasks (MBPP, JSON schema, GSM8K, ARC), but this still covers only a small slice of DLM use cases. Generalising the task-dependent peak finding to other open-ended tasks (instruction following, dialog, multi-turn reasoning) requires task-appropriate functional metrics and additional DLMs.

Sample size and seed dependence. Cluster sample sizes are limited by the number of fail cases per (model, dataset) cell: MBPP has ~ 140 fails per DLM, but GSM8K has fewer than 250 each. A three-seed sensitivity check on LLaDA-MBPP (Appendix I) shows pass/fail counts vary by at most ± 3 samples and the plateau structure is preserved across seeds; the temporal-oscillation phenomenon noted in concurrent DLM work (Wang et al., 2025)

would primarily affect borderline samples, which our seed-stability check is consistent with.

Causal interpretation. Even with the negative steering result, our conclusion that “f15601 is a diagnostic correlate rather than a directly controllable cause under our protocol” is interpretation-dependent. A more decisive test would be a counterfactual experiment that surgically replaces the feature direction with a paired pass-cluster feature direction at the peak step, or a gradient-based attribution that traces logit differences back to the SAE layer. We leave such causal-mechanistic follow-up to future work.

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A Top-Feature Persistence Across Steps

Figure 3 shows the pairwise Jaccard of top-20 fail features between denoising steps and the enrichment trajectory of three illustrative features for LLaDA-8B / MBPP. Table 2 reports the per-step silhouette, permutation p , and top fail-enriched feature. f15601 is the only feature among the top-20 at all three peak-or-later steps (32, 64, 127).

step	best K	silhouette / p	top fail feat
0	2	0.271 / 0.418	f6421 (+0.295)
1	2	0.220 / 0.793	f5561 (+0.313)
4	2	0.231 / 0.727	f5561 (+0.309)
16	2	0.440 / 0.541	f3892 (+0.326)
32	2	0.702 / 0.118	f15601 (+0.391)
64	2	0.662 / 0.033	f15601 (+0.385)
127	2	0.601 / 0.144	f11265 (+0.329)

Table 2: LLaDA-8B / MBPP per-step diagnose summary.

B Qualitative Cluster Inspection

Table 3 shows illustrative LLaDA-MBPP failures from the two step-64 fail clusters. We re-generated the first ten fail samples from each cluster with the baseline LLaDA decoding setup and re-ran the MBPP tests. Cluster 0 yields 3 assertion failures, 2 syntax errors, 4 name errors, and 1 value error; cluster 1 yields 5 assertion failures, 4 name errors, and 1 type error. The examples are not intended as human-validated feature labels; they only show that both sparse clusters contain concrete functional failures, ranging from malformed code to plausible but semantically wrong implementations.

C Step-64 Cross-grid Snapshot

Table 4 summarises silhouette + permutation p across the 4 datasets $\times$ 3 models grid at step 64, complementing the trajectory-shape claim in Appendix E.

dataset	LLaDA	Dream	Qwen
mbpp	0.66 / 0.033	0.50 / 0.128	0.16 / 0.106
json	0.58 / 0.195	0.77 / 0.135	0.22 / 0.007
gsm8k	0.38 / 0.432	0.29 / 0.449	0.15 / 0.315
arc	0.43 / 0.312	0.18 / 0.882	0.13 / 0.343

Table 4: Silhouette / permutation p at step 64 per condition; bold $p < 0.05$. Full trajectory in Table 5.

D Generation and Probing Details

Denoising procedure. Both models run 128 denoising steps. LLaDA uses block-based unmask-

ing: the generation region is divided into blocks of 32 tokens, and each block is denoised sequentially over $128/(\text{gen_length}/32)$ steps per block. Within each block, the model selects the most confident tokens to unmask at each step. Dream uses a global linear schedule: at step i , the fraction of tokens to unmask is determined by a linearly decreasing timestep schedule from $t = 1$ to $t = \epsilon$, and the most confident masked tokens are unmasked globally.

Prompt construction. For JSON schema, we prepend a system prompt containing the target schema and append a JSON code-fence marker as a generation prefix. For GSM8K, the system prompt instructs the model to solve step-by-step and end with ##### followed by the numeric answer. Both models use their respective chat templates. A complete prompt example for each task is shown in Figure 4.

Hidden state extraction. At each of the 7 checkpoint steps $\{0, 1, 4, 16, 32, 64, 127\}$, we run a forward pass with `output_hidden_states=True` and extract the residual stream at the SAE layer. The generation region is partitioned into 4 equal-length position regions and mean-pooled within each region, producing a d -dimensional vector per region (4096 for LLaDA, 3584 for Dream and Qwen). The region-mean is passed through the DLM-Scope SAE for feature analysis (§3).

Steering hook implementation. The steering hook is registered on the SAE-layer transformer block via `register_forward_hook`. The hook receives the tuple output, re-encodes `output[0]` through the SAE in-place, computes $\alpha \cdot z_{i^*} \cdot W_{\text{dec}}[:, i^*]$ and subtracts the contribution. A counter and per-step diagnostic prints confirm the hook fires $T=128$ times per generation (once per denoising step) and that the modification magnitude $\|\Delta h\|$ is non-zero.

Computational resources. All generation, encoding, and steering experiments use NVIDIA A100 (80GB) GPUs on Modal. Models are loaded in bfloat16; SAE weights cast to bfloat16 for the steering hook. Total compute: approximately 40 A100-hours across all reported experiments.

Reproducibility and determinism. Generation uses seed 0; both LLaDA and Dream commit the highest-confidence masked positions deterministically, so the fail/pass partition per cell is repro-

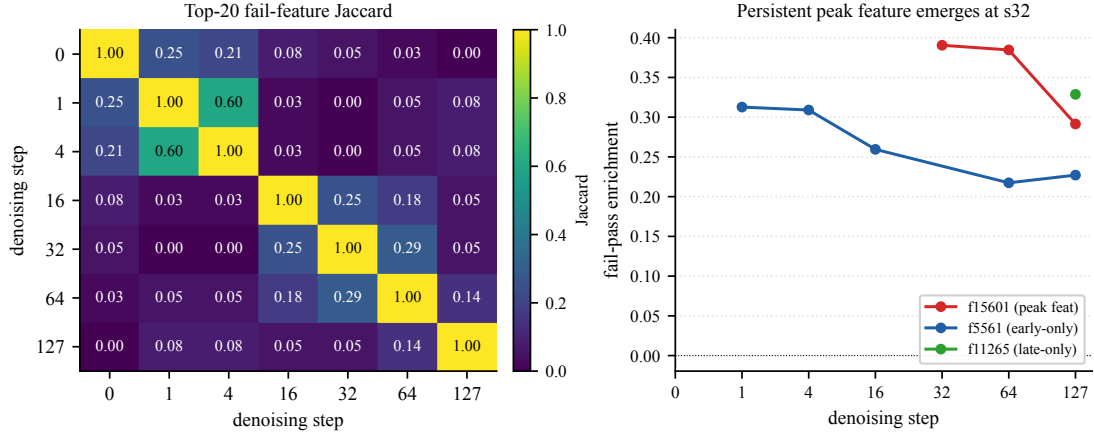


Figure 3: Left: pairwise Jaccard of top-20 fail features between denoising steps. Right: enrichment trajectory of f15601 (persistent peak), f5561 (early only), and f11265 (final only).

cluster	characteristic features	task	observed failure
0	f11404, f9798, f189, f496, f8825	closest smaller number	Generates a long case split that returns n for small inputs instead of $n-1$, failing assertions.
0	f11404, f9798, f189, f496, f8825	unique element in sorted array	Produces malformed pointer-update code with a syntax error.
1	f189, f492, f3320, f11404, f15601	remove first/last occurrence	Removes only boundary characters and misses interior first/last occurrences, failing assertions.
1	f189, f492, f3320, f11404, f15601	lowercase underscore regex	Uses <code>re.compile</code> without importing <code>re</code> , causing a <code>NameError</code> .

Table 3: Illustrative generated failures from LLaDA-MBPP step-64 fail clusters. Error-type counts are computed over 10 regenerated failures per cluster; full generated outputs and errors are saved in `mbpp_llada/cluster_failure_examples.json` on the Modal probe-results volume.

ducible across runs. Hidden states are cached as `bfloat16` tensors; SAE encoding reads from the dense residual cache on demand, so top-20 selection, KMeans, and permutation null are rerunnable without re-invoking either DLM. The permutation

null uses 1000 shuffles per cell with a fixed permutation seed offset, so silhouette p -values are themselves reproducible.

E Full Cross-grid Trajectory

Table 5 reports the signal-to-null gap (KMeans silhouette — permutation null mean) at each of the five denoising checkpoints, for both DLMs $\times$ four tasks. The peak step per cell is bolded; the only DLM peak with $p < 0.05$ is LLaDA-8B / MBPP at step 64. The table separates the trajectory-shape claim from per-step significance: modest gaps can still support the timing pattern when the same task peaks at the same denoising phase across both DLMs.

F Per-condition Trajectory Detail

Table 6 reports per-step silhouette, permutation p -value, and top fail feature for the four (DLM, task) conditions not detailed in Appendix A. Two patterns are worth flagging. Dream-7B / JSON schema concentrates on f6326, which is the top-1 fail-enriched feature at four of five checkpoint

JSON schema (LLaDA chat template)

```
[BOS] system\n You are a helpful assistant
that answers in JSON. Here's the JSON schema
you must adhere to:\n <schema>\n {schema} \n
</schema>\n user\n {user input}\n assistant\n
""json\n
```

GSM8K (Dream chat template)

```
[BOS] im_start system\n Solve the math problem
step by step. End your answer with ####
followed by the final numeric answer. im_end\n
im_start user\n {question} im_end\n im_start
assistant\n
```

Figure 4: Schematic prompt examples after each model’s chat template is applied (special tokens shown as plain words for readability; the actual tokens used are model-specific control tokens). The generation region (128–512 masked tokens) is appended after the final assistant marker. Both models share the same system/user content; only the wrapping tokens differ.

model	task	s=4	s=16	s=32	s=64	s=127	peak
<i>LLaDA-8B</i>							
	mbpp	−0.06	−0.07	+0.18	+0.31	+0.18	s64
	jsonschema	+0.08	+0.06	+0.01	+0.14	+0.12	s64
	gsm8k	−0.01	−0.05	−0.08	−0.00	+0.22	s127
	arc	+0.20	+0.13	+0.14	+0.08	−0.14	s4
<i>Dream-7B</i>							
	mbpp	+0.15	+0.03	+0.02	+0.15	−0.03	s64
	jsonschema	+0.04	+0.06	+0.16	+0.23	+0.17	s64
	gsm8k	+0.05	+0.04	+0.04	+0.01	+0.16	s127
	arc	−0.14	−0.05	−0.02	−0.15	+0.26	s127

Table 5: Signal-to-null gap (observed silhouette − permutation null mean) at five denoising checkpoints across the 2×4 (DLM, task) grid. Bold marks the peak step per cell. Cross-DLM peak-step agreement: MBPP (both s64), JSON (both s64), GSM8K (both s127); only ARC disagrees (LLaDA s4 vs Dream s127).

steps; this mirrors the persistent-feature behaviour seen for f15601 on LLaDA-MBPP and supports the cross-DLM peak-step claim at the feature level. In LLaDA-8B / GSM8K and LLaDA-8B / ARC after step 32 no single feature surfaces as a stable top-1 marker (dashes in the table); the cluster signal there is spread across multiple weakly-enriched features rather than concentrated on one atom, consistent with the negative single-feature steering result in §4.

step	K	silhouette / p	top fail feat
<i>Dream-7B / MBPP</i> (peak s64; below null at s127)			
4	2	0.420 / 0.107	f15414 (+0.272)
16	2	0.413 / 0.231	f6326 (+0.342)
32	2	0.451 / 0.407	f10891 (+0.408)
64	3	0.497 / 0.128	f6326 (+0.421)
127	2	0.333 / 0.460	f15414 (+0.270)
<i>Dream-7B / JSON schema</i> (peak s64, +0.23 gap)			
4	2	0.344 / 0.256	f6326 (+0.427)
16	2	0.401 / 0.213	f10823 (+0.433)
32	2	0.572 / 0.121	f6326 (+0.419)
64	2	0.773 / 0.135	f6326 (+0.420)
127	3	0.475 / 0.109	f233 (+0.349)
<i>LLaDA-8B / GSM8K</i> (late peak; s127 spike +0.22 gap)			
4	3	0.190 / 0.394	f2519 (+0.197)
16	3	0.175 / 0.629	f13085 (+0.192)
32	2	0.255 / 0.661	— (−)
64	2	0.379 / 0.432	— (−)
127	2	0.701 / 0.197	— (+0.224 gap)
<i>LLaDA-8B / ARC</i> (anomalous s4 peak, +0.20 gap)			
4	2	0.627 / 0.079	f4643 (+0.186)
16	2	0.532 / 0.227	f3892 (+0.243)
32	2	0.480 / 0.155	— (+0.142 gap)
64	2	0.427 / 0.312	— (+0.075 gap)
127	2	0.316 / 0.784	— (−0.142 gap)

Table 6: Per-step diagnose for the four (DLM, task) conditions complementing Table 2. Dashes mark no single feature surfaces as the top-1 marker (cluster persists across multiple weakly-enriched features).

G Single-feature vs. Raw-probe AUC

We compare the cross-validated AUC of (i) the single most-fail-enriched SAE feature, (ii) top- N SAE features (logistic regression with per-fold selection), and (iii) a raw-probe baseline (PCA-64 + LR on the same hidden state) across all 12 cells (Table 7). Raw-probe AUC dominates in 11 of 12 cells; the only SAE wins (Qwen / MBPP, Dream / ARC) are cells where the raw probe is itself near chance. Top-20 SAE features recover roughly 85–95% of the raw-probe AUC, suggesting sparse atoms carry most but not all of the signal in the dense residual. This complements the negative steering finding (§4): even at optimal feature selection, sparse linear features under-represent the information that determines correctness.

model	dataset	top-1	top-5	top-20	raw
<i>LLaDA-8B</i>					
	mbpp	0.59	0.73	0.73	0.82
	jsonschema	0.67	0.66	0.69	0.81
	gsm8k	0.69	0.71	0.72	0.77
	arc	0.61	0.65	0.66	0.71
<i>Dream-7B</i>					
	mbpp	0.70	0.69	0.73	0.82
	jsonschema	0.68	0.74	0.78	0.80
	gsm8k	0.57	0.61	0.66	0.70
	arc	0.49	0.54	0.56	0.57
<i>Qwen-2.5-7B (AR baseline)</i>					
	mbpp	0.57	0.66	0.59	0.61
	jsonschema	0.59	0.66	0.71	0.75
	gsm8k	0.59	0.66	0.71	0.75
	arc	0.56	0.59	0.65	0.74

Table 7: Per-cell 5-fold cross-validated AUC at step 64 predicting functional correctness from top- N SAE features (logistic regression, per-fold top- N selection fitted on training folds only) vs. raw-probe baseline (PCA-64 fitted on training folds only + LR, applied to the *same* SAE-layer hidden state). Bold marks the per-cell maximum.

H Dense Temporal Sweep on MBPP

We re-generate MBPP on both DLMs and capture hidden states at 9 dense checkpoints in [48, 80] (every 4 steps) and an additional 6 checkpoints in [84, 124] (every 8 steps), then run the same TopK-SAE encoding, top-20 fail enrichment, KMeans silhouette, and permutation null pipeline as the main 7-point grid. This sweep was requested as a robustness check against sparse-sampling artefacts. Generation uses seed 0 throughout; counts are LLaDA 115/142 and Dream 125/132 pass/fail. Three seeds confirm stability (Appendix D). Table 8 shows per-step silhouette, permutation null mean, gap, p -value, and top-1 fail feature for the [48, 80] plateau region; Appendix I reports the [84, 124] post-plateau region across three seeds.

step	sil	null	gap	p	top-1
<i>LLaDA-8B / MBPP</i>					
48	0.680	0.400	+0.280	0.037	f15601
52	0.728	0.392	+0.336	0.006	f2144
56	0.708	0.376	+0.331	0.012	f15601
60	0.627	0.362	+0.266	0.050	f2144
64	0.609	0.355	+0.253	0.081	f15601
68	0.719	0.350	+0.369	0.005	f15601
72	0.709	0.338	+0.371	0.014	f15601
76	0.710	0.359	+0.351	0.026	f15601
80	0.583	0.364	+0.219	0.121	f15601
<i>Dream-7B / MBPP</i>					
48	0.452	0.347	+0.105	0.152	f10891
52	0.450	0.344	+0.106	0.171	f10891
56	0.476	0.342	+0.134	0.121	f11613
60	0.470	0.355	+0.114	0.164	f14029
64	0.462	0.354	+0.107	0.177	f6326
68	0.506	0.350	+0.156	0.108	f6326
72	0.516	0.442	+0.074	0.252	f6326
76	0.525	0.441	+0.084	0.258	f6326
80	0.495	0.452	+0.043	0.321	f14029

Table 8: Dense step sweep (48–80 every 4 steps) on LLaDA-MBPP and Dream-MBPP. Bold $p < 0.05$. On LLaDA, 7 of 9 sub-steps reach $p < 0.05$ and f15601 is top-1 at 7 of 9 steps, confirming a broad plateau rather than a single-step peak.

Three findings. **Plateau structure.** On LLaDA-MBPP the gap stays above +0.19 throughout [48, 116] with multiple significant sub-steps across three seeds; step 64 (the only $p < 0.05$ point on the sparse 7-grid) sits in the middle of this plateau rather than at a sharp peak. The sparse-grid claim that step 64 is the unique peak should thus be read as “step 64 is a representative point in the plateau”. **Late collapse.** The signal collapses sharply at step 124 (~ 3 steps before the final state), consistent with the residual stream re-organising as the generation completes block-by-block commits. **Plateau feature hand-off.** f15601 (LLaDA) is the top-1 fail-enriched feature at most dense steps in [48, 100]; it is replaced by f3892 at the late-plateau steps 108–116. Dream-MBPP also shows a persis-

tent top-1 (f6326) at [60, 76].

I Seed Sensitivity

We re-run the LLaDA-MBPP dense sweep (Appendix H) with two additional generation seeds (seed 1 and seed 2). Pass/fail counts vary by at most ± 3 samples across seeds (seed 0: 115/142, seed 1: 112/145, seed 2: 116/141). Table 9 reports per-step gap and p -value for each seed and the top-1 fail-enriched feature.

step	seed 0 gap	seed 0 p	seed 1 gap	seed 1 p	seed 2 gap	seed 2 p	top-1 majority
0	+0.20	.02	+0.08	.15	+0.02	.35	mixed
1	+0.04	.15	+0.01	.40	+0.03	.20	f5561/f6421
4	−0.11	.91	−0.09	.84	−0.09	.87	mixed
16	−0.16	.69	−0.05	.54	−0.15	.67	mixed
32	+0.14	.18	+0.20	.10	+0.19	.11	f15601
48	+0.28	.04	+0.31	.01	+0.28	.04	f15601
52	+0.34	.01	+0.31	.02	+0.33	.002	f2144/f15601
56	+0.33	.01	+0.31	.02	+0.33	.01	f15601
60	+0.27	.05	+0.31	.02	+0.32	.02	f2144
64	+0.25	.08	+0.25	.09	+0.28	.05	mixed
68	+0.37	.005	+0.33	.02	+0.34	.03	f15601
72	+0.37	.01	+0.36	.01	+0.37	.02	f15601
76	+0.35	.03	+0.32	.04	+0.34	.04	f15601
80	+0.22	.12	+0.22	.11	+0.25	.13	f15601
84	+0.24	.10	+0.20	.15	+0.24	.12	f15601/f2087
92	+0.23	.08	+0.19	.15	+0.19	.19	f15601/f2087
100	+0.20	.16	+0.28	.09	+0.25	.14	f15601/f2087
108	+0.33	.08	+0.24	.14	+0.36	.05	f3892
116	+0.41	<.001	+0.22	.15	+0.30	.10	f3892
124	+0.00	.46	+0.13	.23	+0.19	.19	f14728/f12454
127	+0.14	.20	+0.14	.20	+0.17	.19	f15601/f11265

Table 9: Full 21-step sweep across three seeds on LLaDA-MBPP (anchors {0, 1, 4, 16, 32}, plateau {48–80}, post-plateau {84–124}, final state 127); all from a single new-prompt generation run so fail/pass sets are identical within each seed. Bold $p < 0.05$. The plateau spans steps 48–116 with ≥ 7 sub-steps significant in each seed; pre-plateau steps 1–16 show no signal and step 4 has negative gap (residual stream below random structure); step 124 collapses just before the final state.

J Intervention Effect Size

The negative steering result in §4 could be artefactual if the intervention failed to meaningfully perturb the SAE-layer residual. We log per-step activation of the target feature (a_{target}), the L2 norm of the steering delta ($\|\Delta h\|$), and its ratio to the hidden state norm ($\|\Delta h\|/\|h\|$) for $n=8$ cluster-1 fail cases under each intervention condition. Table 10 reports averages.

condition	a_{target}	$\ \Delta h\ $	ratio
baseline ($\alpha=0$)	9.47	0.00	0%
suppress f15601 s64	9.43	185	4.4%
suppress f15601 s16	9.27	183	4.4%
suppress top-5 s64	47.4	624	14.8%
reverse f15601 s64	9.43	185	4.4%

Table 10: Effect-size diagnostics on $n=8$ cluster-1 fail samples per condition. All four active conditions induce $\|\Delta h\| \geq 180$ ($\geq 4\%$ of $\|h\|$); the top-5 combined condition reaches 14.8%. Despite measurable perturbation, no condition flips any label.

The intervention is therefore not silent: $\|\Delta h\|$ ranges from 4.4% of $\|h\|$ for single-feature suppression to 14.8% for the top-5 combined condition, and f15601 is reliably active ($a_{\text{target}} \approx 9.4$) during the intervention window in all sampled cases. The null steering result is consistent with functional correctness being distributively encoded such that perturbing a sparse direction at the SAE-layer residual does not propagate into a label flip, even when that perturbation reaches $\sim 15\%$ of the residual norm.

K Post-hoc Selection Sensitivity

We reanalyse cached LLaDA-MBPP plateau residuals (seed 0) at the strongest plateau steps and report three robustness checks (Table 11). **(a) Top- N :** silhouette + permutation across $N \in \{10, 20, 30, 50\}$. Step 68 remains $p < 0.05$ at all four; step 64 degrades for $N \geq 30$, marking step 68 as the more robust plateau point. **(b) Clustering method:** at $N=20$, KMeans, Agglomerative (Ward), and HDBSCAN (min_size=5) all recover $K=2$ with near-identical silhouette. **(c) Labels-hidden:** clustering the full 257-sample fail+pass set at $K=2$ with the same top-20 features (selected via labels, but the cluster itself is unsupervised) gives $\text{ARI} \approx 0$, confirming the silhouette signal measures within-fail sub-structure rather than fail/pass separability.

	step 64				step 68			
	sil	null	gap	p	sil	null	gap	p
<i>(a) Top-N sweep, KMeans best-K</i>								
$N=10$	0.711	0.366	+0.34	.019	0.739	0.363	+0.38	.015
$N=20$	0.609	0.355	+0.25	.081	0.719	0.350	+0.37	.005
$N=30$	0.590	0.385	+0.20	.110	0.665	0.374	+0.29	.016
$N=50$	0.568	0.443	+0.12	.192	0.644	0.416	+0.23	.009
<i>(b) Clustering method, $N=20$</i>								
KMeans	$K=2$, sil = 0.609				$K=2$, sil = 0.719			
Agglo. Ward	$K=2$, sil = 0.609				$K=2$, sil = 0.719			
HDBSCAN	$K=2$, sil = 0.610 (11 noise)				$K=2$, sil = 0.712 (11 noise)			
<i>(c) Labels-hidden: $K=2$ KMeans on full $n=257$ at $N=20$</i>								
ARI / p	−0.007 / 0.865				−0.007 / 0.905			

Table 11: Sensitivity at LLaDA-MBPP plateau steps. (a) Step 68 is robust across N ; (b) three clustering methods agree on $K=2$ and silhouette; (c) labels-hidden $\text{ARI} \approx 0$ confirms the cluster signal is within-fail sub-structure, not fail/pass separability.

L Functional Correctness Definitions

Every task uses a deterministic, machine-checkable rubric; samples with no extractable candidate are labelled fail. **MBPP:** extract the first def-block or fenced “python block, prepend test_imports, run all assertions with a 10-second timeout; syntax errors, timeouts, and assertion failures are fails. **JSON schema:** output must parse as valid JSON and match the reference structure (top-level type, required keys, scalar-leaf types), ignoring key order and extra keys. **GSM8K:** compare only the final numeric answer; extract `####\s*([-\.d.]+)` and compare to gold after stripping commas and trailing zeros. **ARC-Challenge:** extract `####\s*([A-D])`; fallback to the last standalone A-D letter.